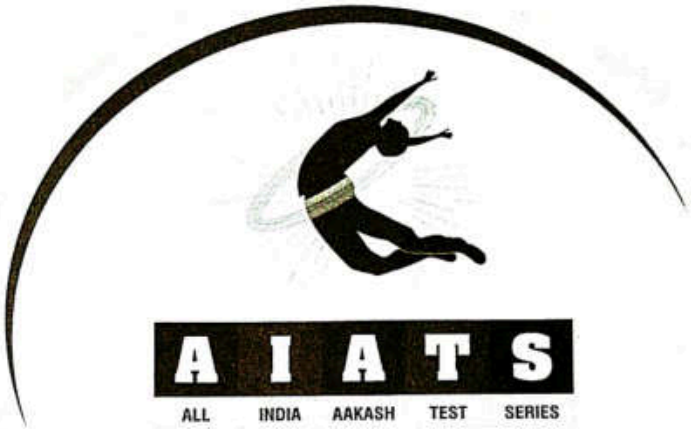
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Test Date: 30-11-2025

Aakash
Medical | IIT-JEE | Foundations

F
Booklet Code

Aakash East Zone



A I A T S
ALL INDIA AAKASH TEST SERIES

for

Medical Entrance Exam - 2026

National Eligibility-cum-Entrance Test (NEET)

TEST No. 2

(XII Passed Students)

INSTRUCTIONS FOR CANDIDATES

1. Read each question carefully.
2. It is mandatory to use Blue/Black Ball Point Pen to darken the appropriate circle in the answer sheet.
3. Mark should be dark and should completely fill the circle.
4. Rough work must not be done on the answer sheet.
5. Do not use white-fluid or any other rubbing material on answer sheet. No change in the answer once marked is allowed.
6. Student cannot use log tables and calculators or any other material in the examination hall.
7. Before attempting the question paper, student should ensure that the test paper contains all pages and no page is missing.
8. Before handing over the answer sheet to the invigilator, candidate should check that Roll No. and Centre Code have been filled and marked correctly.
9. Immediately after the prescribed examination time is over, the answer sheet to be returned to the invigilator.
10. (i) The question paper consists of 180 compulsory questions (45 questions each in Physics and Chemistry and 90 questions in Biology (Botany & Zoology)). The total duration of the examination will be 180 minutes (3 Hrs). (ii) Each correct answer carries four marks. One mark will be deducted for each incorrect answer from the total score.

Note : It is compulsory to fill **Roll No.** and **Test Booklet Code** on answer sheet, otherwise your answer sheet will not be considered.

Test No. 2

TOPICS OF THE TEST

Physics	Laws of Motion; Work, Energy and Power
Chemistry	Classification of Elements and Periodicity in properties, Chemical Bonding and Molecular Structure
Botany	Biological Classification, Morphology of Flowering Plants
Zoology	Breathing and Exchange of Gases, Body Fluids and Circulation

MM : 720

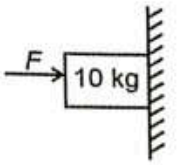
TEST - 2

Time : 3 Hrs.

[PHYSICS]

Choose the correct answer :

1. If the minimum force $F = 200$ N is required to hold a 10 kg block against a wall as shown in figure, then the coefficient of friction between the block and wall is



(1) $\frac{3}{4}$ (2) $\frac{1}{3}$

(3) $\frac{1}{4}$ (4) $\frac{1}{2}$

2. A car of mass 1000 kg is moving on a level circular track of radius 80 m. If coefficient of static friction between the road and tyre of car is 0.5, the maximum speed of the car during the turn can be

(1) 10 m/s (2) 30 m/s

(3) 20 m/s (4) 40 m/s

3. A ball of mass m is hanging vertically by a light inextensible string of length L from a fixed-point O. The ball is given a speed $u = \sqrt{4gL}$ at the lowest position. The height reached by the ball from lowest position to the point, where string slacks, will be

(1) $\frac{7}{5}L$ (2) $\frac{5L}{3}$

(3) $\frac{L}{6}$ (4) $\frac{7}{3}L$

4. **Statement A:** The force of friction only acts between the two surfaces when one surface is in relative motion w.r.t. the other.

Statement B: The phenomenon of raising outer edge of a curved road above the inner edge is called banking of the road.

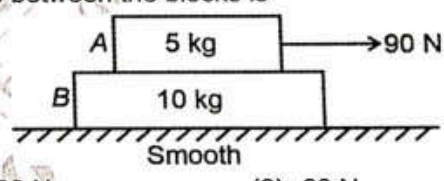
(1) Both statements A and B are correct

(2) Both statements A and B are incorrect

(3) Statement A is correct and statement B is incorrect

(4) Statement A is incorrect and statement B is correct

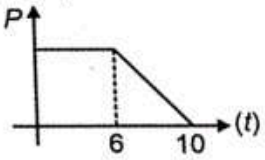
5. A block A of mass 5 kg is placed on a block B of mass 10 kg which is placed on a smooth surface. A force of 90 N is applied on the upper block as shown in figure. If both the blocks move together, then friction force between the blocks is



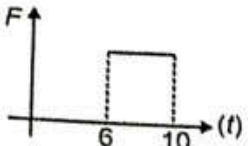
(1) 30 N (2) 60 N

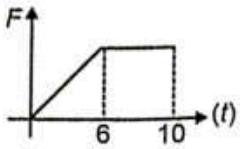
(3) 20 N (4) 80 N

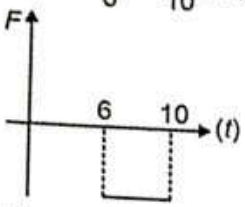
6. The momentum (P) of an object varies with time (t) as shown in figure. The corresponding force (F) versus time (t) graph will be

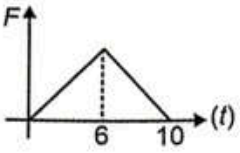


Space for Rough Work

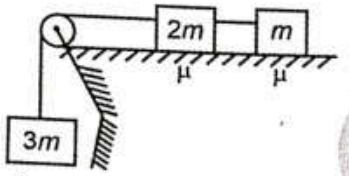
(1) 

(2) 

(3) 

(4) 

7. A system consists of three masses $3m$, $2m$ and m connected by a string passing over a pulley. Mass $3m$ hangs freely while $2m$ and m are placed on the rough horizontal table (the coefficient of friction = μ). The pulley is frictionless and has negligible mass. If the system is released from rest, the acceleration of $3m$ will be



(1) $\frac{g(1-\mu)}{3}$

(2) $\frac{g(1-2\mu)}{3}$

(3) $\frac{g(1-\mu)}{2}$

(4) $\frac{g(1-2\mu)}{2}$

8. If impulse I varies with time t as I(kg m/s) = $(10t^2 - 4t)$. The change in momentum is zero at

(1) $t = \frac{1}{2}$ s

(2) $t = \frac{1}{4}$ s

(3) $t = \frac{2}{5}$ s

(4) $t = \frac{1}{10}$ s

9. A turn of radius 100 m is banked for an optimum speed of 20 m s^{-1}. If a vehicle of mass 500 kg negotiates the curve at a speed of 15 m s^{-1}, then the force of friction on it will be nearly (Assume that friction is sufficient to prevent skidding and slipping and take tan22° = 0.4 and cos22° = 0.93)

(1) 13.75 N

(2) 813.75 N

(3) 572.75 N

(4) 72.75 N

10. A particle is moving in a circle of radius R = 4 m with angular velocity ω = 2 rad/s. At t = 0 it starts accelerating with an angular acceleration α = 1 rad/s^2. The angle between the tangential acceleration and the net acceleration at t = 2 s is

(1) $\tan^{-1}(8)$

(2) $\tan^{-1}(32)$

(3) $\tan^{-1}(16)$

(4) $\tan^{-1}(64)$

11. A machine gun fires bullets each of mass 50 g with velocity of 100 m/s. The man holding the gun can exert a maximum force of 40 N on it. How many bullets can he fire per second at most?

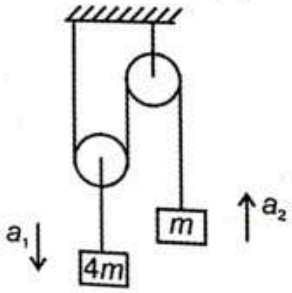
(1) 4

(2) 12

(3) 6

(4) 8

12. In the given figure, if the system is released from rest, then relation between acceleration of mass $4m$ (a_1) and acceleration of mass m (a_2) will be



(1) $2a_1 = a_2$

(2) $a_1 = 2a_2$

(3) $a_1 = a_2$

(4) $a_1 = 4a_2$

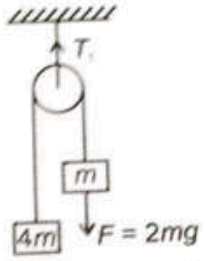
Space for Rough Work

13. If thrust force on a rocket, that releases the exhaust gases with a relative velocity of 180 m/s is 2000 N, then the rate of combustion of the fuel is

(1) $\frac{100}{9}$ kg/s (2) $\frac{100}{7}$ kg/s

(3) $\frac{100}{13}$ kg/s (4) $\frac{100}{3}$ kg/s

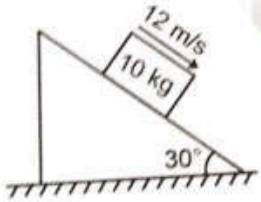
14. In the figure shown, string is massless and pulley is smooth then the tension T_1 in the string will be



(1) $\frac{16mg}{5}$ (2) $\frac{32mg}{5}$

(3) $\frac{5mg}{2}$ (4) $\frac{9mg}{4}$

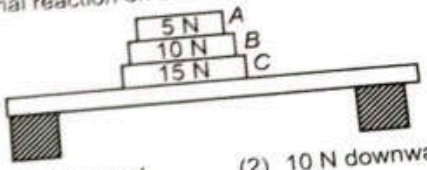
15. A block of mass 10 kg is moving on a fixed inclined plane with constant velocity 12 m/s as shown in figure. The coefficient of kinetic friction between the inclined plane and the block is



(1) $\frac{1}{\sqrt{2}}$ (2) $\frac{1}{\sqrt{3}}$

(3) 0.75 (4) 0.25

16. Three books (A, B and C) rest on a table. The weight of each book is indicated as shown below. The normal reaction on book B by book C is



(1) 10 N upward (2) 10 N downward

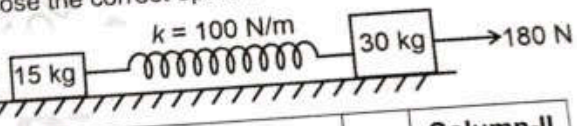
(3) 15 N upward (4) 15 N downward

17. A block of mass m is attached with a massless string of breaking strength $3mg$. If block is moving up with the help of string, then the maximum acceleration of the block will be

(1) g (2) $3g$

(3) $4g$ (4) $2g$

18. In the system shown in figure, the mass 30 kg is getting pulled by a force 180 N and at the given instant 15 kg mass has acceleration 5 m/s^2 towards right. On the basis of given information some physical quantities are asked in column-I and their value in SI unit is given in column-II. Match the physical quantity with its corresponding value and choose the correct option.



	Column-I		Column-II
a.	Acceleration of 30 kg block	(i)	3.5
b.	Extension in the spring	(ii)	105
c.	Net force on 30 kg block	(iii)	75
d.	Value of spring force	(iv)	$\frac{3}{4}$

(1) a(i), b(iv), c(ii), d(iii) (2) a(iv), b(i), c(iii), d(ii)

(3) a(i), b(ii), c(iv), d(iii) (4) a(ii), b(iii), c(i), d(iv)

Space for Rough Work

19. A block of mass m is placed on a smooth wedge of inclination angle θ from horizontal. The whole system is accelerated horizontally with acceleration a, so that block does not slip on the wedge. The normal reaction on the block by the wedge will be

(1) $\frac{mg}{\tan\theta}$ (2) $\frac{mg}{\cos\theta}$

(3) $mg\tan\theta$ (4) $mg\sin\theta$

20. Which of the following option is correct for a body moving with constant velocity?

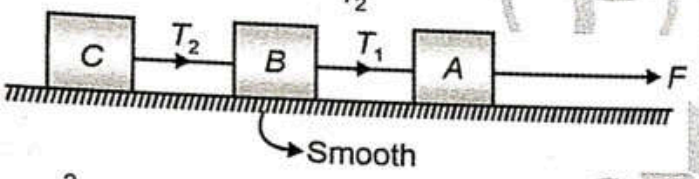
(1) There is net force acting on the body

(2) Net force acting on the body is zero

(3) Linear momentum of the body is zero

(4) Linear momentum of the body is increasing continuously

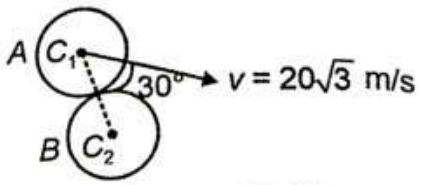
21. Three blocks A, B and C having masses $2m$, m and $3m$ respectively, connected with ideal light strings. If a constant force F is applied on block A as shown in figure, then the value of $\frac{T_1}{T_2}$ will be



(1) $\frac{3}{2}$ (2) $\frac{4}{3}$

(3) $\frac{1}{2}$ (4) $\frac{1}{5}$

22. A ball A of mass 0.1 kg collides elastically with another identical ball B (at rest) with velocity $20\sqrt{3}$ m/s at angle 30° from the line joining their centres C_1 and C_2 as shown in figure. The kinetic energy of the ball B just after the collision will be (All surfaces are smooth)



(1) 60 J (2) 120 J

(3) 90 J (4) 45 J

23. Consider the statements with regards to non-conservative force and choose the correct option.

(1) It has no potential energy associated with it

(2) The work done by it is dependent on path of the body

(3) When the starting and end points are same, the total work is not necessarily zero

(4) All of the above

24. A bullet hits a block at rest on a frictionless horizontal surface and gets embedded into it. The physical quantity associated with the (bullet + block) system, that remains unchanged is

(1) Kinetic energy (2) Temperature

(3) Linear momentum (4) Both (1) & (2)

25. A spring of force constant k is stretched by a small length x. The work done in stretching it further by a small distance y is

(1) $\frac{1}{2}ky(y+2x)$ (2) $\frac{1}{2}k(x+y)^2$

(3) $\frac{1}{2}k(x+2y)^2$ (4) $\frac{1}{2}k(y^2-x^2)$

26. An object is in equilibrium. If on slight displacement of the object from equilibrium position and when it is released, it comes back to its original position, then this equilibrium of object is called

(1) Neutral equilibrium (2) Unstable equilibrium

(3) Stable equilibrium (4) Both (1) and (2)

Space for Rough Work

27. **Assertion (A):** A spring has potential energy when it is compressed or stretched.

Reason (R): In compressing or stretching, work is done on the spring against the restoring force.

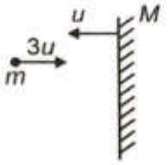
(1) Both Assertion and Reason are true and Reason is the correct explanation of Assertion

(2) Both Assertion and Reason are true but Reason is not the correct explanation of Assertion

(3) Assertion is true but Reason is false

(4) Both Assertion and Reason are false

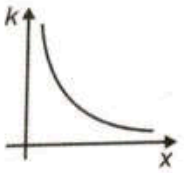
28. A ball of mass m moving with speed $3u$ collides with a very heavy wall of mass M ($M >> m$) moving with speed u as shown in figure. The speed of the ball just after the collision will be (Consider collision is perfectly elastic)

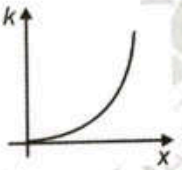


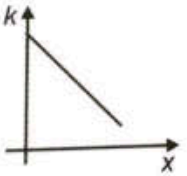
(1) $2u$ (2) $3u$

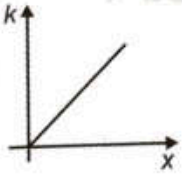
(3) $5u$ (4) $4u$

29. Choose the correct graph of kinetic energy(k) of a freely falling body versus distance(x) travelled by the body.

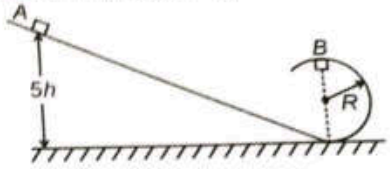
(1) 

(2) 

(3) 

(4) 

30. In the figure shown, a small block of mass m is released from point A on a smooth track. When the block reaches point B, then normal force applied by the track on it will be ($R = h$)



(1) $7mg$ (2) $5mg$

(3) $3mg$ (4) $6mg$

31. A body of mass 4 kg moves according to relation $x = (2t^2 + t^3)$ where x is in m and t is in second, the change in kinetic energy of the body in time interval $t = 1$ s to $t = 2$ s will be

(1) 302 J (2) 702 J

(3) 151 J (4) 351 J

32. When two bodies of same mass undergoes head-on elastic collision, then

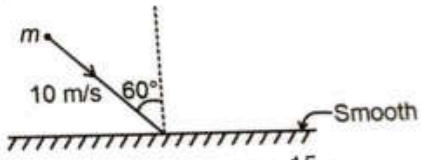
(1) Their velocities are interchanged

(2) Their kinetic energy gets interchanged

(3) Their linear momentum gets interchanged

(4) All of these

33. A ball of mass $m = 250$ g collides with smooth ground at an angle 60° with the vertical. If initial speed of ball is 10 m/s and the collision lasts for time 0.5 s, then average force exerted by the ground on the ball will be (consider coefficient of restitution $e = \frac{1}{2}$)



(1) $\frac{5}{2}$ N (2) $\frac{15}{2}$ N

(3) $\frac{15}{4}$ N (4) 5 N

Space for Rough Work

34. The power of a machine varies with time (t) given as $P = (3t^2 + 2t + 1)$ W, where t is in second. The work done by the machine in time interval $t = 1$ s to $t = 3$ s will be

(1) 18 J (2) 36 J
(3) 12 J (4) 24 J

35. An object is displaced from position A (1, 2) m to position B (3, 4) m under a force $\vec{F} = (2x\,\hat{i} + 2\hat{j})$ N, the work done by this force will be

(1) 12 J (2) 8 J
(3) 16 J (4) 4 J

36. Consider the following statements and choose the correct option.

(a) Potential energy can be positive, negative or zero.

(b) A moving body may or may not have potential energy.

(c) Potential energy is not defined for non-conservative force.

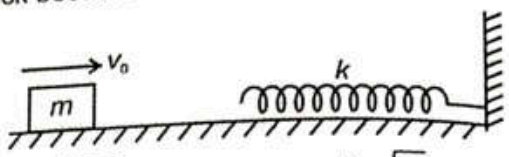
(1) Only (b) is correct
(2) Only (b) and (c) are correct
(3) Only (a) and (b) are correct
(4) All (a), (b) and (c) are correct

37. **Assertion (A):** Power developed in uniform circular motion is always zero.

Reason (R): In uniform circular motion, force and velocity are perpendicular to each other.

(1) Both Assertion and Reason are true and Reason is the correct explanation of Assertion
(2) Both Assertion and Reason are true but Reason is not the correct explanation of Assertion
(3) Assertion is true but Reason is false
(4) Both Assertion and Reason are false

38. A block is moving with velocity v_0 on a smooth horizontal ground and strikes a spring in its natural length having spring constant k as shown in figure. The compression in the spring when velocity of the block becomes half, will be

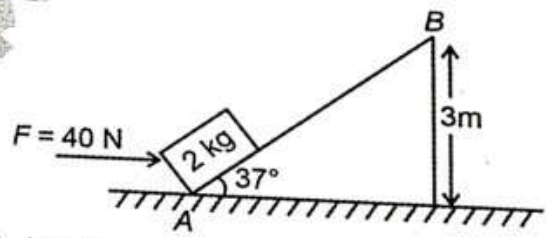


(1) $\frac{v_0}{2}\sqrt{\frac{3m}{k}}$ (2) $\frac{v_0}{2}\sqrt{\frac{m}{k}}$
(3) $v_0\sqrt{\frac{3m}{k}}$ (4) $v_0\sqrt{\frac{3m}{2k}}$

39. A body of mass m is projected upward with an initial speed $\sqrt{2gh}$, if it returns to ground with a speed $\sqrt{\frac{2gh}{3}}$, then the magnitude of work done by the air resistance will be

(1) $\frac{mgh}{3}$ (2) $\frac{mgh}{2}$
(3) $\frac{3mgh}{4}$ (4) $\frac{2mgh}{3}$

40. A constant horizontal force $F = 40$ N starts acting on a block of mass 2 kg from position A as shown in figure. The velocity of the block when it reaches to the top position B of the fixed and smooth wedge, will be



(1) 15 m/s (2) 10 m/s
(3) 4 m/s (4) 8 m/s

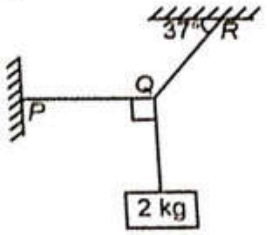
Space for Rough Work

41. The work done by the force $\vec{F} = (2\hat{i} + 2\hat{j} - \hat{k})$ N when it displaces the body from a point (2, 3, – 4) m to a point (5, 1, 3) m is
(1) 10 J (2) –5 J
(3) –9 J (4) 17 J

42. **Statement A:** The collision in which the bodies stick together after collision is called perfectly elastic collision.
Statement B: In perfectly inelastic collision, linear momentum of system is conserved.
Choose the **correct** option.
(1) Both statements A and B are correct
(2) Statement A is correct and statement B is incorrect
(3) Both statements A and B are incorrect
(4) Statement A is incorrect and statement B is correct

43. A body is moved along a straight line by a machine delivering constant power. The kinetic energy of the body is proportional to time (t) as
(1) $t^{\frac{1}{2}}$ (2) $t^{\frac{3}{4}}$
(3) t (4) t^2

44. A 2 kg block is kept in equilibrium with the help of strings as shown in the figure. The value of tension in string QR is



(1) 25 N (2) $\frac{75}{2}$ N
(3) $\frac{80}{3}$ N (4) $\frac{100}{3}$ N

45. If $\vec{P} + \vec{Q} = \vec{R}$, then magnitude of $\vec{Q}$ is
(1) $R - P$
(2) $P + R$
(3) $\sqrt{\vec{R}.\vec{Q} - \vec{P}.\vec{Q}}$
(4) $\sqrt{\vec{R}.\vec{P} - \vec{P}.\vec{Q}}$

[CHEMISTRY]

46. Dobereiner's Triads among the following are
(a) Li, Na, K (b) Ca, Sr, Ba
(c) C, N, O (d) Cl, Br, I
(1) (a) and (b) only
(2) (b), (c) and (d) only
(3) (a), (b) and (d) only
(4) (a) and (c) only

47. Eka-Aluminium and Eka-Silicon respectively are
(1) Ga and Ge (2) Ge and Ga
(3) Sc and Ga (4) Ga and Sc

48. Given below are two statements:
Statement I: According to Mendeleev's periodic law, the properties of the elements are periodic functions of their atomic numbers.
Statement II: According to Modern periodic law, the properties of the elements are a periodic function of their atomic weights.
In the light of the above statements, choose the correct answer.
(1) Statement I is correct but statement II is incorrect
(2) Statement I is incorrect but statement II is correct
(3) Both statement I and statement II are correct
(4) Both statement I and statement II are incorrect

Space for Rough Work

49. Consider the following statements
(a) Group 3 of long form of periodic table consists maximum number of elements.
(b) Seventh period of long form of periodic table can accommodate maximum 49 elements.
(c) Group 17 contains most electronegative element of periodic table.
(d) 3rd and 4th period consist equal number of elements.
(e) First lanthanoid element is introduced in seventh period.
Correct statements are
(1) (a), (b) and (c) only (2) (d) and (e) only
(3) (b), (d) and (e) only (4) (a) and (c) only

50. The element with atomic number 113 belongs to
(1) 7th period and 13th group
(2) 6th period and 13th group
(3) 6th period and 15th group
(4) 7th period and 15th group

51. Given below are two statements:
Statement I: Co, Rh and Ir belong to 9th group in modern periodic table.
Statement II: Element with atomic number 39 is the first element in which electron is introduced in 4d subshell.
In the light of the above statements, choose the correct answer.
(1) Statement I is correct but statement II is incorrect
(2) Statement I is incorrect but statement II is correct
(3) Both statement I and statement II are correct
(4) Both statement I and statement II are incorrect

52. Which among the following is transuranium element?
(1) Th (2) Pa
(3) Np (4) Lu

53. If an ion M^{2+} has electronic configuration $[Ar]3d^{10}4s^0$, then the element M belongs to
(1) 4th period and 14th group
(2) 5th period and 14th group
(3) 5th period and 15th group
(4) 4th period and 12th group

54. Match atomic number in List-I with their IUPAC official name in List-II.

	List-I		List-II
a.	101	(i)	Bohrium
b.	103	(ii)	Mendelevium
c.	105	(iii)	Lawrencium
d.	107	(iv)	Dubnium

The correct match is
(1) a(iii), b(ii), c(iv), d(i) (2) a(ii), b(iii), c(i), d(iv)
(3) a(ii), b(iii), c(iv), d(i) (4) a(ii), b(iv), c(iii), d(i)

55. Given below are two statements:
Statement I: IUPAC name of the element with atomic number 124 is Unbiquadium.
Statement II: Element with atomic number 120 will be present in eighth period of long from of periodic table.
In the light of the above statements, choose the correct answer.
(1) Statement I is correct but statement II is incorrect
(2) Statement I is incorrect but statement II is correct
(3) Both statement I and statement II are correct
(4) Both statement I and statement II are incorrect

56. Match elements in List-I with their classification in List-II.

	List-I		List-II
a.	Antimony	(i)	Transition metal
b.	Rhenium	(ii)	Non-metal
c.	Promethium	(iii)	Lanthanoid
d.	Chlorine	(iv)	Semi-metal

The correct match is
(1) a(iv), b(ii), c(i), d(iii) (2) a(iii), b(i), c(iv), d(ii)
(3) a(iv), b(iii), c(i), d(ii) (4) a(iv), b(i), c(iii), d(ii)

57. Which among the following orders of atomic radii is not correct?
(1) Li < Na < K < Rb (2) F < Cl < Br < I
(3) C < Be < B < Li (4) Cl < S < P < Si

58. Given below are two statements:
Statement I: Ionisation enthalpy of any element is equal to magnitude of electron gain enthalpy of its unipositive cation.
Statement II: Noble gases have large positive electron gain enthalpy.
In the light of the above statements, choose the correct answer.
(1) Statement I is correct but statement II is incorrect
(2) Statement I is incorrect but statement II is correct
(3) Both statement I and statement II are correct
(4) Both statement I and statement II are incorrect

59. Consider the following statements
(a) For isoelectronic species, the cation with the greater positive charge will have a larger radius.
(b) $O^-(g) + e^- \longrightarrow O^{2-}(g)$; is an endothermic process.
(c) During formation of Sc to Sc^+, electron is removed from 3d subshell.
(d) Ionisation enthalpy of $Al^+(g)$ will be more than that of Al(g).
The correct statements are
(1) (a) and (b) only (2) (a), (b) and (d) only
(3) (b) and (d) only (4) (b) and (c) only

60. Most electronegative chalcogen is
(1) F (2) S
(3) N (4) O

61. The correct order of negative electron gain enthalpy of following elements is
(1) O > S > Se > Te (2) S > O > Se > Te
(3) S > Se > Te > O (4) S > Se > O > Te

62. Correct order of first ionisation enthalpy is
(1) Be < B < C < O < N (2) B < Be < C < O < N
(3) B < Be < C < N < O (4) B < Be < O < C < N

63. The ionic radius of Na^+ ion is 1.02 Å. The ionic radii (in Å) of Mg^{2+} and Al^{3+}, respectively are
(1) 0.68 and 0.72 (2) 0.85 and 0.99
(3) 1.05 and 0.99 (4) 0.72 and 0.54

64. Match successive ionisation enthalpy in given List-I with electronic configuration given in List-II.

	List-I (Ionization Energy in kJ mol^{-1})		List-II (Electronic configuration)
a.	x, $2x$, $5x$, $125x$, $130x$	(i)	$1s^2, 2s^2, 2p^6, 3s^1$
b.	y, $30y$, $33y$, $37y$, $43y$	(ii)	$1s^2, 2s^2, 2p^6, 3s^2$
c.	z, $5z$, $90z$, $94z$, $99z$	(iii)	$1s^2, 2s^2, 2p^6, 3s^2, 3p^1$
d.	p, $4p$, $9p$, $17p$, $102p$	(iv)	$1s^2, 2s^2, 2p^6, 3s^2, 3p^2$

Correct match is
(1) a(i), b(iii), c(ii), d(iv) (2) a(i), b(iii), c(iv), d(ii)
(3) a(iii), b(i), c(ii), d(iv) (4) a(iii), b(i), c(iv), d(ii)

65. Match type of hydride of element 'E' given in List-I with its group number given in List-II.

	List-I		List-II
a.	EH_2	(i)	15
b.	EH_4	(ii)	2
c.	EH_3	(iii)	1
d.	EH	(iv)	14

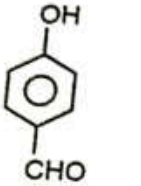
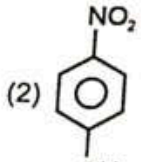
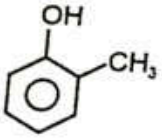
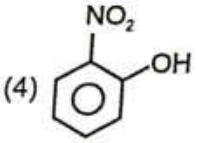
Correct match is
(1) a(ii), b(iv), c(i), d(iii) (2) a(ii), b(iv), c(iii), d(i)
(3) a(iv), b(ii), c(i), d(iii) (4) a(iv), b(ii), c(iii), d(i)

Space for Rough Work

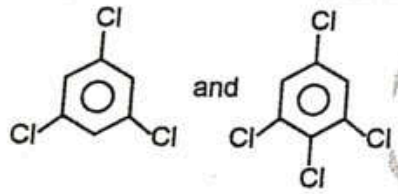
66. Which of the following block of periodic table consist metal, non-metal and metalloid together?

(1) *s*-block (2) *p*-block
(3) *d*-block (4) *f*-block

67. Which among the following molecule will exhibit intramolecular H-bonding?

(1) 
(2) 
(3) 
(4) 

68. If dipole moment of chlorobenzene is x, then dipole moment of given molecules respectively will be



(1) x and zero (2) 3x and 4x
(3) Zero and x (4) x and 2x

69. Consider the following molecules on the basis of molecular orbital theory.

(a) N_2 (b) O_2
(c) B_2 (d) C_2
(e) Li_2

Molecule(s) with more than one π bond is/are

(1) (a) only
(2) (a) and (d) only
(3) (a) and (b) only
(4) (c), (d) and (e) only

70. Given below are two statements:

Statement I: O_2 is a paramagnetic molecule with magnetic moment of 1.73 BM.

Statement II: During conversion of N_2 to N_2^+ electron is removed from $\sigma 2p_z$ orbital.

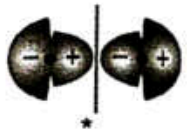
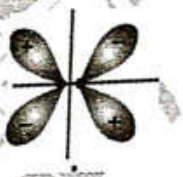
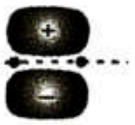
In the light of the above statements, choose the correct answer.

(1) Statement I is correct but statement II is incorrect
(2) Statement I is incorrect but statement II is correct
(3) Both statement I and statement II are correct
(4) Both statement I and statement II are incorrect

71. Observed dipole moment of A – B bond is 1.88 D and A – B bond length is 99 pm. The percentage ionic character in A – B bond is

(1) 60.43 (2) 39.57
(3) 74.24 (4) 25.76

72. Which among the following correctly represents a π antibonding molecular orbital?

(1) 
(2) 
(3) 
(4) 

73. Consider the following statements one is labelled as **Assertion A** and the other is labelled as **Reason R:**

Assertion A: During formation of molecular orbital of homo atomic species by the combination of atomic orbitals, 1s orbital cannot combine with 2s orbital.

Reason R: The combining atomic orbitals must have the same or nearly the same energy.

In the light of the above statements, choose the **correct** answer from the options given below:

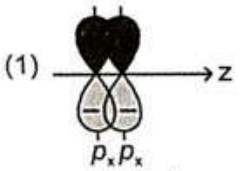
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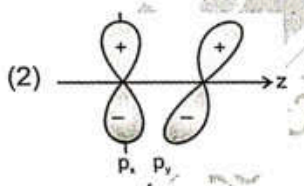
(1) Both **A** and **R** are true and **R** is the correct explanation of **A**
(2) Both **A** and **R** are true but **R** is not the correct explanation of **A**
(3) **A** is true but **R** is false
(4) **A** is false but **R** is true

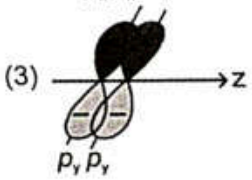
74. *d*-orbital involved in dsp^2 hybridisation is
(1) d_{z^2} (2) d_{xy}
(3) d_{y^2} (4) $d_{x^2-y^2}$

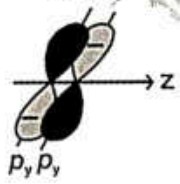
75. Central atom of which among the following species is not sp^3 hybridised?
(1) NH_3 (2) ClO_4^-
(3) PCl_3 (4) BrF_3

76. Which among the following is not a salient feature of hybridisation?
(1) The number of hybrid orbitals is equal to the number of the atomic orbitals that get hybridised.
(2) The hybridised orbitals are always equivalent in energy and not equivalent in shape.
(3) The hybrid orbitals are more effective in forming stable bonds than the pure atomic orbitals.
(4) These hybrid orbitals are directed in space in some preferred direction to have minimum repulsion between electron pairs.

77. Which among the following is considered as zero overlapping?
(1) 

(2) 

(3) 

(4) 

78. Which among the following statement about PCl_5 is incorrect?
(1) Number of lone pairs present is 15
(2) Hybridisation and shape are sp^3d and trigonal bipyramidal respectively
(3) Axial bonds are shorter than equatorial bonds
(4) It is a non-polar and non-planar molecule

79. Match the molecule in List-I with the shape given in List-II.

	List-I		List-II
a.	SO_2	(i)	Bent
b.	ClF_3	(ii)	Square planar
c.	SF_4	(iii)	See-saw
d.	XeF_4	(iv)	T-shape

Correct match is
(1) a(iii), b(ii), c(i), d(iv) (2) a(i), b(ii), c(iii), d(iv)
(3) a(iii), b(iv), c(i), d(ii) (4) a(i), b(iv), c(iii), d(ii)

80. Correct order of repulsive interaction of electron pairs is
(1) Lone pair (lp) – Lone pair (lp) > Lone pair (lp) – Bond pair (bp) > Bond pair (bp) – Bond pair (bp)
(2) Bond pair (bp) – Bond pair (bp) > Lone pair (lp) – Lone pair (lp) > Lone pair (lp) – Bond pair (bp)
(3) Bond pair (bp) – Bond pair (bp) > Lone pair (lp) – Bond pair (bp) > Lone pair (lp) – Lone pair (lp)
(4) Lone pair (lp) – Bond pair (bp) > Lone pair (lp) – Lone pair (lp) > Bond pair (bp) – Bond pair (bp)

81. Correct order of dipole moment is
(1) $H_2O > HCl > NH_3$ (2) $H_2O > NH_3 > HCl$
(3) $NH_3 > H_2O > HCl$ (4) $HCl > H_2O > NH_3$

82. Which among the following statements is incorrect?
(1) The cannonical forms have real existence
(2) The molecule does not exist for a certain fraction of time in one cannonical form and for other fractions of time in other cannonical forms.
(3) There is no such equilibrium between the cannonical forms as we have between tautomeric forms (keto and enol) in tautomerism.
(4) The molecule as such has a single structure which is the resonance hybrid of the cannonical forms and which cannot as such be depicted by a single Lewis structure.

Space for Rough Work

83. Consider the following statements

(a) Isoelectronic species with no resonance have identical bond order.

(b) NO is an odd electron species.

(c) O_3 exists in three resonating structures.

(d) H_2 molecule has less bond enthalpy as compared to HCl molecule.

(e) Resonance averages the bond characteristics as a whole.

Correct statements are

(1) (b), (c) and (d) only

(2) (c), (d) and (e) only

(3) (a), (b) and (e) only

(4) (a), (b), (d) and (e) only

84. Given below are two statements:

Statement I: For given processes

$H_2O(g) \rightarrow H(g) + OH(g)$, $\Delta_a H_1 = a$ kJ mol^{-1}

$OH(g) \rightarrow H(g) + O(g)$, $\Delta_a H_2 = b$ kJ mol^{-1}

$a < b$

Statement II: Second O – H bond in H_2O molecule is stronger than the first O – H bond.

In the light of the above statements, choose the correct answer.

(1) Statement I is correct but statement II is incorrect

(2) Statement I is incorrect but statement II is correct

(3) Both statement I and statement II are correct

(4) Both statement I and statement II are incorrect

85. Correct order of covalent bond length is

(1) N – O < C – H < C – O

(2) C – H < N – O < C – O

(3) C – H < C – O < N – O

(4) C – O < C – H < N – O

86. Correct order of Lattice energies for the given compounds is

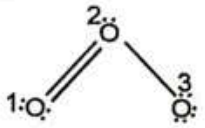
(1) $Na_2O < MgO < Al_2O_3$

(2) $Al_2O_3 < MgO < Na_2O$

(3) $Na_2O < Al_2O_3 < MgO$

(4) $MgO < Na_2O < Al_2O_3$

87. Formal charge on oxygen atoms 1, 2 and 3 respectively in given Lewis dot structure is



(1) 0, +1, –1 (2) –1, 0, +1

(3) –1, +1, 0 (4) 0, –1, +1

88. Which among the underlined elements in given compounds is not correctly matched with its electrovalency?

(1) $Na_2\underline{O}$: 1 (2) $\underline{Al}Cl_3$: 3

(3) $\underline{K}NO_3$: 1 (4) $\underline{Ba}Cl_2$: 2

89. Given below are two statements:

Statement I: H_2SO_4 is an example of expanded octet.

Statement II: Octet of central atom is incomplete in BH_3 molecule.

In the light of the above statements, choose the correct answer.

(1) Statement I is correct but statement II is incorrect

(2) Statement I is incorrect but statement II is correct

(3) Both statement I and statement II are correct

(4) Both statement I and statement II are incorrect

90. Which among the following oxides is not correctly matched with its chemical nature?

(1) Al_2O_3 – Amphoteric (2) N_2O – Acidic

(3) Na_2O – Basic (4) SiO_2 – Acidic

Space for Rough Work

[BOTANY]

91. Select the organism which is devoid of flagella.
(1) *Trypanosoma* (2) *Gonyaulax*
(3) *Plasmodium* (4) *Euglena*

92. Consider the following statements and select the **correct** option.

Statement I: In members of phycomycetes, asexual reproduction takes place by zoospores or aplanospores.

Statement II: According to R.H. Whittaker's classification, bladderwort, *Cuscuta* and Venus fly trap are placed under kingdom plantae in spite of the fact that they are partial heterotrophs.

In the light of above statements choose the most appropriate option.
(1) Only statement I is correct
(2) Only statement II is correct
(3) Both statement I and statement II are correct
(4) Both statement I and statement II are incorrect

93. In general, the viruses that infect plants have the genetic material as
(1) ssRNA (2) dsRNA
(3) ssDNA (4) dsDNA

94. The organism which causes sleeping sickness belongs to the group in which the members
(1) Have a cavity that opens to the outside of the cell surface
(2) Are either free living or parasitic
(3) Have an infectious spore like stage
(4) Have silica shell on their surface

95. Red tides are caused due to rapid multiplication of
(1) Euglenoids (2) *Nostoc*
(3) *Gonyaulax* (4) Chrysophytes

96. Choose the **Incorrect** statement for Mycoplasma.
(1) It does not have cell wall
(2) It can survive without oxygen
(3) It is pathogenic only to plants
(4) It is the smallest living cell known

97. Select the bacteria which are present in the gut of several ruminant animals and are responsible for production of biogas from the dung of cattle.
(1) Methanogens (2) Thermoacidophiles
(3) Halophiles (4) Cyanobacteria

98. According to R.H. Whittaker's five kingdom classification, organisms with simple body organization but most complex behaviour is placed under the kingdom
(1) Fungi (2) Plantae
(3) Protista (4) Monera

99. Lichens are the mutualistic relationship between algae and fungi in which
(1) Mycobiont shows autotrophic mode of nutrition
(2) Phycobiont absorbs mineral nutrients from substratum
(3) Phycobiont can fix atmospheric nitrogen
(4) Mycobiont takes shelter from phycobiont

100. Which among the following features is **not** common in class ascomycetes and basidiomycetes?
(1) An intervening dikaryotic stage during sexual reproduction occur
(2) Reductional division for the formation of spores in fruiting bodies
(3) Production of asexual spores exogenously
(4) Karyogamy occurs after plasmogamy

101. Select the pair of fungi which are edible and belong to two different classes of fungi
(1) Mushrooms and morels
(2) Morels and truffles
(3) *Puccinia* and *Agaricus*
(4) *Claviceps* and *Ustilago*

Space for Rough Work

102. Viroids differ from prions as the
(1) Former lack protein coat and latter have abnormally folded proteins
(2) Former are similar in size to virus and latter are smaller than viruses
(3) Latter have DNA and former have RNA
(4) Latter cause PSTD and former cause mad cow disease

103. Choose the **incorrect** statement among the following.
(1) Stigma is the receptive surface for the pollen grains
(2) An ovary can have more than one ovule
(3) Ovary is the part of carpel
(4) Lotus and pea have single carpel

104. Choose the ornamental plant of Fabaceae family.
(1) Tulip (2) Ashwagandha
(3) *Gloriosa* (4) Lupin

105. In three domain system of classification, which kingdom is divided into two domains?
(1) Protista (2) Monera
(3) Fungi (4) Plantae

106. Given below are two statements of Assertion (A) and Reason (R).

Assertion (A): In slime moulds, under unfavourable condition, the plasmodium differentiates and forms the fruiting bodies having spores.

Reason (R): The spores produced by slime moulds are extremely resistant and can survive for many years under adverse condition.

In the light of above statements, choose the **most appropriate** answer.
(1) Both (A) and (R) are true and (R) is the correct explanation of (A)
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)
(3) (A) is true but (R) is false
(4) (A) is false but (R) is true

107. Match the following Column I with Column II and choose the **correct** option.

	Column I		Column II
A.	M.W. Beijerinck	(i)	He showed that acellular organisms could be crystallized and crystals consist largely of proteins
B.	W.M. Stanley	(ii)	He discovered infectious RNA particles which are devoid of protein coat
C.	Dmitri Ivanowsky	(iii)	He demonstrated that the extract of the infected plant of tobacco could cause infection in healthy plants
D.	T.O. Diener	(iv)	He recognized certain microbes as a causal organism of the mosaic disease of tobacco

	A	B	C	D
(1)	(iii)	(i)	(iv)	(ii)
(2)	(iv)	(iii)	(i)	(ii)
(3)	(ii)	(i)	(iv)	(iii)
(4)	(iii)	(i)	(ii)	(iv)

108. Match Column I with Column II and choose the **correct** option.

	Column I		Column II
A.	*Albugo*	(i)	Sexual spores are produced endogenously
B.	*Agaricus*	(ii)	They are commonly known as the imperfect fungi
C.	*Colletotrichum*	(iii)	The hyphae in them are continuous tubes filled with multinucleated cytoplasm
D.	*Neurospora*	(iv)	Fragmentation is the common mode of vegetative reproduction

Space for Rough Work

	A	B	C	D
(1)	(iii)	(iv)	(i)	(ii)
(2)	(ii)	(iii)	(iv)	(i)
(3)	(iv)	(ii)	(i)	(iii)
(4)	(iii)	(iv)	(ii)	(i)

109. The classification system given by Linnaeus was able to distinguish between the organisms having
(1) Unicellular or multicellular body organisation
(2) Photosynthetic or non-photosynthetic nutrition
(3) Cell wall or without cell wall
(4) Eukaryotic or prokaryotic cell type

110. Select the **incorrectly** matched pair from the following

(1)	Basidium is the site of karyogamy and meiosis	–	*Puccinia*
(2)	Zygospore is formed by fusion of two gametes	–	*Neurospora*
(3)	Only asexual or vegetative phase of these fungi are known	–	*Alternaria*
(4)	A unicellular sac fungus	–	*Saccharomyces cerevisiae*

111. Read the below given features
(i) They have branched and septate mycelium.
(ii) Asexual spores are generally absent.
(iii) Sexual reproduction involves the formation of the dikaryotic phase.
(iv) The sexual spores are produced exogenously.
(v) They can be infectious to plants.
Select the group of fungi in which all the given features are observed.
(1) Ascomycetes (2) Basidiomycetes
(3) Deuteromycetes (4) Zygomycetes

112. Consider the following steps which are involved in the sexual cycle of fungi
(A) Fusion of two nuclei.
(B) Fusion of protoplasm between two motile or non-motile gametes.
(C) Meiosis in zygote results in haploid spore production.
Choose the **correct** chronological order of the sequence of events which are taking place in the sexual life cycle of the fungi.
(1) B→C→A (2) C→A→B
(3) A→B→C (4) B→A→C

113. A kingdom 'X' includes only unicellular eukaryotes. It includes certain group of organisms that lack cell wall.
Choose the statements that are true for those certain group of organisms.
(a) Some of them have two unequal sized flagella.
(b) They can move and capture their prey by putting out pseudopodia.
(c) They can have photosynthetic pigments identical to higher plants and show anoxygenic photosynthesis.
(d) They cannot be saprophytic in nature.
(e) Some of them have infectious spore like stage in their life cycle which has staggering effect on human population
Choose the **correct** answer from the options given below.
(1) (a), (b) and (d)
(2) (a), (b) and (e)
(3) (c), (d) and (e)
(4) (b) and (e) only

Space for Rough Work

114. Choose the statement which is **not** true about cyanobacteria.
(1) They have a chlorophyll pigment which is similar to green plants.
(2) They have the ability to fix carbon dioxide and nitrogen in the same cell.
(3) They often form bloom in the polluted water bodies.
(4) Cyanobacteria are unicellular, colonial or filamentous, freshwater/marine or terrestrial algae.

115. Regarding five-kingdom classification system, given by R.H. Whittaker, choose the **correct** one.
(1) Kingdom that includes prokaryotic organisms show only heterotrophic mode of nutrition.
(2) The kingdom of organisms having the most complex body organization show heterotrophic mode of nutrition only.
(3) A kingdom consisting of organisms with a eukaryotic cell type and unicellular body organization, also has presence of chitin in their cell walls.
(4) Kingdom Protista brought together *Chlamydomonas, Chlorella* with *Penicillium* and *Amoeba*.

116. In which among the given plants, a single ovule is attached to its placenta?
(1) Sunflower (2) Makoi
(3) *Dianthus* (4) Tomato

117. In a floral formula, diadelphous condition can be represented as
(1) A_n (2) $A_{(n)+(n)}$
(3) $A_{(n+2)+(n)+(n)}$ (4) $A_{(n)}$

118. Select the plant in which axile placentation is **not** present.
(a) *Nicotiana* (b) Chilli
(c) Lemon (d) *Argemone*
(e) Tomato (f) *Petunia*
(g) *Dianthus* (h) Mustard
(1) (b), (d), (f) and (g) (2) (a), (c), (e) and (f)
(3) (c), (e) and (f) (4) (d), (g) and (h)

119. Select the **incorrect** statement for cymose inflorescence.
(1) The main axis terminates into a flower
(2) The oldest flower is present at the base
(3) It has limited growth
(4) The flowers are borne in the basipetal order

120. Select the **incorrectly** matched pair from the following.

(1)	Venus fly trap	–	The leaves get modified to trap insects for nutrition
(2)	*Calotropis*	–	A pair of leaves at each node on opposite side
(3)	Pea	–	Shows basal placentation
(4)	*Alstonia*	–	More than two leaves arise at each node

121. Consider the following statements and select the **correct** option.
Statement I: In a floral diagram, calyx, corolla, androecium and gynoecium are drawn in successive whorls, calyx being the outermost and the gynoecium being in the centre.
Statement II: In China rose, cohesion of stamens is not observed.
In the light of above statements choose the **most appropriate** option.
(1) Statement II is true but statement I is false
(2) Statement II is false but statement I is true
(3) Both statement I and statement II are true
(4) Both statement I and statement II are false

Space for Rough Work

122. Match the following columns and select the **correct** option.

	Column I		Column II
a.	Petiole	(i)	It is the green expanded part of the leaf with vein and veinlets
b.	Leaf blade	(ii)	It is the swollen region in some leguminous plants where leaf is attached to the stem
c.	Leaf base	(iii)	This structure allows the lamina to flutter in the wind
d.	Pulvinus	(iv)	In monocots, this structure expands into a sheath covering the stem partially or wholly

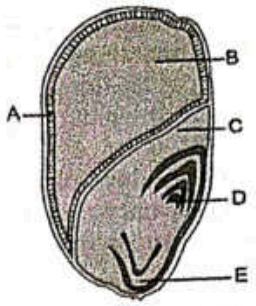
(1) a(iv), b(i), c(iii), d(ii) (2) a(iii), b(i), c(ii), d(iv)
(3) a(ii), b(iv), c(i), d(iii) (4) a(iii), b(i), c(iv), d(ii)

123. Lily flower differs from bean flower as in the former
(1) Perianth is absent
(2) All four floral whorls are found
(3) Sepal is absent
(4) Anther is absent

124. Consider the following features of a plant
(a) Flowers can be divided into two equal halves by any plane passing through the centre.
(b) The margin of sepals or petals just touch one another but without overlapping.
(c) Stamens are found to be attached to the sepals of the flower.
(d) Gynoecium is bicarpellary and placenta is swollen with many ovules.
(e) Seeds are mainly non-endospermous
(f) Ovules develop on central axis and septa are absent.

How many of the above features are true for the members of the potato family?
(1) Four (2) Five
(3) Three (4) Two

125. Consider the given diagram and choose the **correct** option



(1) Label A represents the lipid rich layer which is the covering of the embryo of the seed
(2) B is the result of double fertilization
(3) Label D and E are enclosed in a sheath known as coleorhiza and coleoptile respectively
(4) Structure labelled as C is found in both monocot and dicot seeds

126. Region of root which is responsible for the growth of the root in length
(1) Is called region of elongation
(2) Has epidermal hairs which are very fine and delicate
(3) Is responsible for the absorption of water and minerals from the soil
(4) Has small thick-walled cells which are able to divide repeatedly

127. Read the following statements
Statement I: Leaf-like structures present at the base of petiole are called bract.
Statement II: Axillary buds later develop into a branch or a leaf.
In the light of above statements choose the **correct** option.
(1) Statement I is correct but statement II is incorrect
(2) Statement II is correct but statement I is incorrect
(3) Both statement I and statement II are correct
(4) Both statement I and statement II are incorrect

Space for Rough Work

128. Select the **correct** statement for silk cotton plant.
(1) Pinnately compound leaves are found
(2) Number of leaflets are present on rachis
(3) The tip of the petiole bears all the leaflets in the form of bunch or cluster
(4) Leaves are with entire lamina without any incision

129. Read the following statements:
Assertion (A): A typical flower has four different kind of whorls arranged successively on the swollen end of the pedicel.
Reason (R): When shoot apical meristem changes to floral meristem then the internode get swollen and axis gets condensed.
In the light of above statements choose the **most appropriate** answer.
(1) Both (A) and (R) are true and (R) is correct explanation of (A)
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)
(3) (A) is true but (R) is false
(4) (A) is false but (R) is true

130. In monocotyledonous plant, such as wheat
(1) The radicle leads to the formation of the fibrous roots which are short lived
(2) The leaf base gets expanded into a sheath called pulvinus
(3) Embryo is made up of radicle, embryonal axis and two cotyledons
(4) The seeds obtained are endospermous

131. Select the **incorrectly** matched pair

(1)	Perianth	–	Found in flowers where calyx and corolla are indistinguishable
(2)	Unisexual flower	–	Presence of either stamen or carpel
(3)	Cruciform corolla	–	It is found in the flower of mustard
(4)	Epicarp	–	Fruit wall that is differentiated into three layers

132. Buds are **not** found at/in
(a) Terminal position of stem
(b) Axil of compound leaf
(c) Nodes
(d) Axil of leaflets
(e) Internodes of stem
(f) Underground part of stem
(1) (a), (c) and (e) (2) (d) and (e)
(3) (b), (c) and (d) (4) Only (a) and (d)

133. Select the feature(s) which are **true** for aleurone layer.
(a) Surrounds nutritive tissue
(b) It is the innermost covering of endosperm
(c) Found in the monocot seed
(d) Proteinaceous in nature
(e) Surrounds the embryo
(1) (a), (c) and (e)
(2) All (a), (b), (c), (d) and (e)
(3) (a), (c) and (d)
(4) (d) and (e)

134. In vexillary aestivation, there are five petals in which
(1) Two lateral petals are known as wings
(2) The anterior one is known as standard
(3) Two petals are attached with androecium
(4) Two posterior petals form the keel

135. Which among the following plant parts lack chlorophyll in it?
(1) Sepal of flower
(2) Petiole of leaf
(3) Dormant axillary bud
(4) Stipule found at the leaf base

Space for Rough Work

[ZOOLOGY]

136. A person 'X', met with an accident and required immediate blood transfusion. The doctor mentions that 'X' is a Rh-positive individual with anti-A antibody in his blood plasma. He can receive blood from all of the following blood group individuals, **except**
(1) O^{-ve} (2) B^{-ve}
(3) A^{+ve} (4) B^{+ve}

137. Select the **incorrect** representation.
(1) FRC = ERV + RV (2) ERV = VC – IC
(3) RV = VC – TV (4) IRV = TLC – (EC + RV)

138. Which of the following is **not** a function performed by conducting part of respiratory system?
(1) Facilitating gaseous exchange between blood and atmospheric air
(2) Humidifying the inhaled air
(3) Clearing inhaled air from foreign particles
(4) Transport of atmospheric air to the alveoli

139. pH of arterial blood is usually
(a) More than that of venous blood
(b) Lesser than that of venous blood
(c) Same as that of venous blood
Choose the **correct** option.
(1) (a) only (2) (b) only
(3) Either (b) or (c) (4) (c) only

140. Consider the given statements w.r.t humans.
(a) Both blood and lymph are fluid connective tissues.
(b) The blood pumped out by each ventricle during a cardiac cycle in case of a healthy adult human is 70 mL under normal conditions.
(c) Blood vascular system consists of a muscular chambered heart, a network of blood vessels and blood.
(d) Chordae tendineae are fibrous cords present in atria and ventricles of the heart.
Select the option representing the **correct** statements only.
(1) (a), (b) and (d) (2) (b), (c) and (d)
(3) (a), (b) and (c) (4) (a), (b), (c) and (d)

141. In a hypothetical condition, if AV node from a human heart is removed but nodal tissues of atrium and ventricles remain intact, it will result in
(1) No change in rhythmic activity of heart
(2) Simultaneous contraction of atria and ventricles
(3) Complete cessation of heart beat
(4) Increase in cardiac output of the individual

142. Select the **correct** statement.
(1) All vertebrates respire through lungs only
(2) All arthropods respire through gills only
(3) All mammals respire via lungs
(4) All fishes and adult amphibians respire via gills

143. 'X' divides into left and right 'Y' bronchi in human respiratory system.
Identify 'X' and 'Y' respectively and choose the **correct** option.
(1) Larynx; Primary (2) Pharynx; Secondary
(3) Trachea; Primary (4) Pharynx; Trachea

144. Select the **correct** option w.r.t. first heart sound 'lub'.
(1) It is produced due to closure of atrio-ventricular valves during ventricular systole.
(2) It is produced due to closure of semilunar valves during ventricular diastole.
(3) It is produced due to closure of atrioventricular valves during ventricular diastole.
(4) It is produced due to opening of semilunar valves during atrial systole.

Space for Rough Work

145. The value of pO_2 in systemic arteries is
(1) 95 mm Hg (2) 45 mm Hg
(3) 104 mm Hg (4) 40 mm Hg

146. Consider the following features.
(a) Generally short lived
(b) Colourless
(c) Nucleated
(d) Number averages 6000-8000 mm^{-3} of blood
All of the above given features are **true** for
(1) Leucocytes (2) Erythrocytes
(3) Thrombocytes (4) Platelets

147. Which among the following is **not** associated with cartilage?
(1) Larynx (2) Secondary bronchi
(3) Epiglottis (4) Alveoli

148. Choose the **incorrect** match w.r.t. approximate values of given aspects in a healthy man.

(1)	Percentage of lymphocytes in total WBCs	Percentage of CO_2 transported as carbamino-hemoglobin
(2)	Percentage of eosinophils in total WBCs	Percentage of O_2 transported in dissolved state
(3)	Percentage of monocytes in total WBCs	Percentage of CO_2 almost transported in dissolved state
(4)	Percentage of basophils in total WBCs	Percentage of O_2 transported as oxyhaemoglobin

149. 'X' is the least abundant of the total WBCs. Identify 'X' and select the **incorrect** option w.r.t. them:
(1) Secrete a vasodilator
(2) Involved in inflammatory response
(3) Are agranulocytes
(4) Secrete an anti-coagulant

150. Match column I with column II.

	Column I		Column II
a.	Cardiac arrest	(i)	Heart is not pumping blood effectively enough to meet needs of the body
b.	Asthma	(ii)	Heart stops beating
c.	Heart failure	(iii)	Difficulty in breathing due to inflammation of bronchi and bronchioles
d.	Emphysema	(iv)	Alveolar walls are damaged due to which respiratory surface is decreased.

Choose the **correct** option.
(1) a(i), b(iii), c(ii), d(iv) (2) a(ii), b(iii), c(i), d(iv)
(3) a(i), b(iv), c(iii), d(ii) (4) a(iii), b(iv), c(i), d(ii)

151. Consider the given statements.
Statement A: All factors in our body are favourable for diffusion of O_2 from alveoli to tissues and that of CO_2 from tissues to alveoli.
Statement B: The valves in our heart ensure that blood flows in a unidirectional manner, preventing back flow *i.e.*, from ventricles to atria.
Select the **correct** option.
(1) Both statements A and B are correct
(2) Both statements A and B are incorrect
(3) Only statement A is correct
(4) Only statement B is correct

152. In adult humans, the thoracic chamber is formed ____ by the vertebral column, ____ by the sternum, and ____ by the ribs.
Fill in the respective blanks by selecting the **correct** option.
(1) Laterally; Dorsally; Ventrally
(2) Ventrally; Laterally; Dorsally
(3) Dorsally; Ventrally; Laterally
(4) Laterally; Ventrally; Dorsally

Space for Rough Work

153. Which of the options given below correctly represents the reaction taking place at tissue site in presence of carbonic anhydrase under normal physiological conditions?

(1) $HCO_3^- + H^+ \rightarrow H_2CO_3 \rightarrow CO_2 + H_2O$

(2) $CO_2 + H_2O \rightarrow H_2CO_3 \rightarrow HCO_3^- + H^+$

(3) $H_2CO_3 \rightarrow HCO_3^- + H^+ \rightarrow CO_2 + H_2O$

(4) $CO_2 + H_2O \rightarrow HCO_3^- + H^+ \rightarrow H_2CO_3$

154 Under normal physiological conditions, oxygen is 50 per cent saturated with haemoglobin, at pO_2 of about

(1) 50 – 55 mm Hg (2) 25 – 30 mm Hg

(3) 15 – 20 mm Hg (4) 65 – 70 mm Hg

155. An _____ blood group individual is considered as universal donor while an _____ blood group individual is considered as universal recipient.

Fill in the respective blanks by selecting the **correct** option.

(1) O^{+ve}; AB^{+ve} (2) O^{-ve}; AB^{-ve}

(3) O^{+ve}; AB^{-ve} (4) O^{-ve}; AB^{+ve}

156. Consider a hypothetical scenario, where cardiac output of an individual is 4 L/minute. If blood volume in left ventricle at the end of ventricular diastole is 90 mL and at end of ventricular systole is 40 mL, then, what is the heart beat rate of this person?

(1) 90 beats per minute (2) 70 beats per minute

(3) 80 beats per minute (4) 40 beats per minute

157. Complete the analogy by selecting the **correct** option.

Complete double circulation : Birds : :

Single circulation : _____

(1) Mammals (2) Reptiles

(3) Amphibians (4) Fishes

158. The centres which regulate cardiac activity and respiratory rhythm are located in

(1) Pons region of forebrain

(2) Cerebellum of hindbrain

(3) Cerebrum of forebrain

(4) Medulla oblongata of hindbrain

159. Match column I with column II w.r.t the blood vessel / structure and the respective chamber it opens into under normal conditions.

	Column I		Column II
a.	Inferior vena cava	(i)	Right ventricle
b.	Tricuspid valve	(ii)	Left atrium
c.	Bicuspid valve	(iii)	Right atrium
d.	Pulmonary vein	(iv)	Left ventricle

Choose the **correct** answer from the options given below.

(1) a(iii), b(iv), c(i), d(ii) (2) a(iii), b(i), c(iv), d(ii)

(3) a(ii), b(i), c(iv), d(iii) (4) a(ii), b(iv), c(i), d(iii)

160. Consider the given statements.

(a) Thrombins are formed from inactive substances present in lymph called fibrinogen.

(b) An injury or trauma stimulates erythrocytes in blood to release certain factors which activate mechanism of coagulation.

(c) Calcium ions play a very important role in clotting.

(d) A clot or coagulum is formed mainly of network of threads in which only damaged thrombocytes are trapped.

How many of the above statements is/are **incorrect**?

(1) One (2) Two

(3) Three (4) Four

Space for Rough Work

161. Select the **Incorrect** option w.r.t. hypertension.
(1) Systolic pressure after repeated checks ≥ 140 mm Hg
(2) Diastolic pressure after repeated checks ≥ 90 mm Hg
(3) Measured by using an electrocardiogram
(4) Can affect brain and kidney and can also lead to heart disease

162. In humans, neural signals through ANS can
(1) Only increase the rate of heart beat
(2) Only decrease the rate of heart beat
(3) Either increase or decrease the rate of heart beat depending upon the neurotransmitter released
(4) Only decrease the speed of conduction of action potentials through cardiac muscle fibres

163. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A): Low pO_2, high pCO_2, high pH and low temperature are factors responsible for dissociation of oxygen from oxyhaemoglobin.

Reason (R): When carbon dioxide is produced by tissues, it dissolves in blood to increase the pH that increases affinity of haemoglobin for oxygen and shifts the O_2 dissociation curve towards right side.

In the light of the above statements, choose the **correct** answer from the options given below :
(1) Both (A) and (R) are true and (R) is the correct explanation of (A)
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)
(3) (A) is true but (R) is false
(4) Both (A) and (R) are false

164. If due to some injury the chordae tendineae attached to the tricuspid valve of human heart become non-functional, what will be the immediate effect?
(1) The pacemaker will stop working
(2) The blood will tend to flow back into the left atrium
(3) The flow of blood into aorta will be increased
(4) The flow of blood into pulmonary artery will be reduced

165. If during an experiment, a drop, each of serum, lymph, plasma and blood were placed on separate slides, which of them will not coagulate after sometimes?
(1) Serum (2) Lymph
(3) Plasma (4) Blood

166. Respiratory minute volume, in case of a healthy adult human can be expressed as
(1) Residual volume × Number of breaths per hour
(2) Tidal volume × Number of breaths per minute
(3) Inspiratory capacity × Number of breaths per minute
(4) Expiratory capacity × Number of breaths per hour

167. Consider the following steps of respiration.
(a) Transport of gases by blood
(b) Pulmonary ventilation
(c) Cellular respiration
(d) Diffusion of O_2 and CO_2 between blood and tissues
(e) Diffusion of gases across alveolar membrane.

Arrange the above given steps in correct order w.r.t. a normal adult human and select the **correct** option.
(1) (e) → (d) → (c) → (b) → (a)
(2) (a) → (c) → (b) → (e) → (d)
(3) (e) → (b) → (a) → (c) → (d)
(4) (b) → (e) → (a) → (d) → (c)

Space for Rough Work

168. In a healthy adult human under resting conditions, at the end of joint diastole, ventricles are
(1) Completely filled with blood
(2) About 1/3rd filled with blood
(3) About 2/3rd filled with blood
(4) Totally empty

169. Relaxation of diaphragm and external inter-costal muscles would cause
(1) Return of ribs and sternum to their original positions
(2) Raising up of ribs along the dorsoventral axis
(3) Increase in intra-pulmonary volume
(4) Decrease in intra-pulmonary pressure

170. Consider the following
(a) Superior vena cava (b) Pulmonary artery
(c) Inferior vena cava (d) Aorta
(e) Pulmonary veins (f) Left ventricle
(g) Right ventricle
Select the option that **correctly** classifies the above mentioned structures based on whether they are part of pulmonary or systemic circulation.

	Pulmonary circulation	Systemic circulation
(1)	(b), (d), (e) and (f)	(a), (c) and (g)
(2)	(a), (b), (c) and (f)	(d), (e) and (g)
(3)	(c), (e) and (g)	(a), (b), (d) and (f)
(4)	(b), (e) and (g)	(a), (c), (d) and (f)

171. If 'A' represents pCO_2 in deoxygenated blood, 'B' represents pO_2 in oxygenated blood and 'C' represents pO_2 in alveoli. Then
(1) A = B = C (2) A > B > C
(3) A < B < C (4) A = B > C

172. Vital capacity does not include
(1) Residual volume
(2) Expiratory reserve volume
(3) Inspiratory reserve volume
(4) Tidal volume

173. Given below are two statements : one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A): Like arteries and veins, capillaries also lack tunica media and tunica externa.

Reason (R): The arteries and veins are responsible for transportation of blood whereas capillaries are permeable for various substances and participate in exchange between blood and tissue fluid.

In the light of the above statements, choose the **correct** answer from the options given below :
(1) (A) is false, (R) is true
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)
(3) Both (A) and (R) are true and (R) is the correct explanation of (A)
(4) Both (A) and (R) are false

174. Select the **correct** statement w.r.t. lymph.
(1) It is devoid of larger proteins and most of the formed elements.
(2) The lymphatic system collects interstitial fluid and drains it back into major arteries.
(3) It is pale yellow coloured fluid containing negligible number of lymphocytes.
(4) It enables intracellular communication due to its hydrophilic nature.

Space for Rough Work

175. Consider the following statements w.r.t. haemoglobin.
(a) Haemoglobin is a red coloured respiratory pigment found in arthropods.
(b) O_2 can bind with haemoglobin in a reversible manner to form oxyhaemoglobin.
(c) Each Hb molecule contains 4 iron atoms which can bind to maximum of 4 oxygen atoms each.
(d) It exhibits quaternary level of protein structure.
How many of the above statements are correct and incorrect, respectively?
(1) 1, 3 (2) 3, 1
(3) 4, 0 (4) 2, 2

176. Receptors associated with aortic arch and carotid artery recognise changes mainly in which among the following and then send necessary signals to the rhythm centre for remedial action?
(a) pCO_2
(b) H^+ concentration
(c) pO_2
Select the **correct** option.
(1) (a) and (b) only (2) (a) only
(3) (b) only (4) (a), (b) and (c)

177. The total thickness of diffusion membrane in human respiratory system is much
(1) More than 2 centimetres
(2) Less than a millimetre
(3) More than a centimetre
(4) More than 2 millimetres

178. Consider the given statements w.r.t. an adult man.
Statement A: Nearly 70 per cent of carbon dioxide is transported as bicarbonate through the blood plasma.
Statement B: The first step in breathing is cellular respiration which involves drawing in of atmospheric air and releasing out of alveolar air.
Select the **correct** option.
(1) Both statements A and B are correct
(2) Both statements A and B are incorrect
(3) Only statement A is correct
(4) Only statement B is correct

179. Complete the analogy by selecting the **correct** option.
Closed circulatory system : Earthworm :: Open circulatory system : _____
(1) Butterfly
(2) Sponges
(3) Humans
(4) Crocodiles

180. Mechanism of breathing vary among different groups of animals depending on
(1) Their habitats only
(2) The levels of body organisation only
(3) Their habitats and levels of body organisation
(4) Only the surface area of their entire body

❑ ❑ ❑

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